

SHRESTH JAISWAL

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EDUCATION

B.Tech in Computer Science and Engineering — Vellore Institute of Technology

Sep 2023 – Present

CGPA: 8.40 / 10.0

SKILLS

Languages: Java, Python, JavaScript, SQL

AI/ML: LLM APIs, AI Output Evaluation

Testing & Automation: Pytest, Selenium, API Testing, Unit Testing, Integration Testing

Backend: Django, Node.js, Express.js, REST APIs

Databases: PostgreSQL, SQL

DevOps & Tools: Git, GitHub Actions, Docker, Linux

Frontend: React.js, Redux Toolkit, React Hook Form, HTML, CSS, TailwindCSS

EXPERIENCE

Software Engineering Intern — Scramble Labs Technology Pvt. Ltd.

May 2025 – Aug 2025

- **Quality Assurance & Testing:** Authored 15 automated unit and integration test suites for new REST APIs, expanding backend code coverage and preventing regression defects prior to production rollout.
- **Git Workflow Optimization:** Established a standardized Git branching and code-review protocol across the engineering team, mitigating 15 potential merge conflicts and boosting team velocity.
- **Scalable Backend Architecture:** Constructed and deployed 2 RESTful APIs using Python/Django and PostgreSQL, implementing strict request validation and query optimization for high-throughput service performance.
- **Frontend Performance Optimization:** Developed 4 modular React.js + TypeScript components within a production Next.js application, reducing page load latency by 15% through React component caching and memoized render hooks.

PROJECTS

TestPilot — AI-Powered Test Automation Platform

Python, FastAPI, Pytest, Selenium, PostgreSQL, LLM API, Docker

- Engineered an AI-driven test generation pipeline that converts API specifications into executable test cases, covering functional, negative, and edge-case scenarios.
- Automated API and UI validation using Pytest and Selenium, with reusable test workflows for authentication, response schemas, status codes, and critical user journeys.
- Built an LLM evaluation framework to assess AI-generated test cases across correctness, consistency, coverage, and practical usability, enabling measurable quality scoring.
- Containerized and productionized the testing platform with Docker, PostgreSQL, and GitHub Actions for reproducible environments, automated test execution, and centralized test-result tracking.

AssetFlow ERP — Enterprise Asset & Lifecycle Management Platform

PostgreSQL, Express.js, React.js, Node.js, Knex.js, REST API | [Code](#) | [Live Demo](#)

- Needed an enterprise-grade platform to manage equipment tracking, checkout workflows, and maintenance requests across teams, with strict, role-appropriate access control.
- Architected the system with 4 granular role-based permission tiers and designed a PostgreSQL relational schema spanning 10+ normalized entities with targeted indexing.
- Built client-side caching (SWR) with asynchronous parallel API execution in React, and enforced transaction audit logging with payload validation across Express.js/Knex.js controllers.
- Achieved a 3x speedup on complex query execution, sub-second rendering across 9 dashboard sections, and full historical data integrity.

Typemate — Rich Text Editor with Smart Autocomplete

React.js, JavaScript, HTML, CSS | [Code](#) | [Live Demo](#)

- Set out to build a dependency-free rich text editor supporting real-time formatting, full accessibility, and faster typing without relying on a third-party editor library.
- Engineered a document editing engine on native Selection and Range APIs with 10+ formatting controls, live word/character counts, and custom undo/redo history.
- Implemented full keyboard accessibility across 20+ toolbar controls using semantic ARIA attributes and roving tab-index navigation, and built dictionary-based prefix-matching autocomplete for word- and sentence-level suggestions.
- Delivered a fully accessible, dependency-free editor with local-storage draft auto-saving that preserves user text across session reloads.

ACHIEVEMENTS & LEADERSHIP

- **Data Structures & Algorithms:** Solved 240+ Data Structures & Algorithms problems on LeetCode.
- **Community Leadership:** Appointed Core Member of Notion Community at VIT Bhopal, organizing tech talks, peer learning workshops, and cross-functional student initiatives.
- **Professional Certifications:** Earned NPTEL Silver Elite Certification in Cloud Computing & Internet of Things (IoT).